BANet: A Balance Attention Network for Anchor-Free Ship Detection in SAR Images

Qi Hu, Shaohai Hu, and Shuaiqi Liu

***Abstract*— Recently, methods based on deep learning have been successfully applied to ship detection for synthetic aperture radar (SAR) images. However, most current ship detection networks rely too much on the anchor mechanism. These methods have low accuracy and poor generalization ability for multiscale ship detection. To solve the aforementioned problems, an anchor-free framework for multiscale ship detection in SAR images based on a balance attention network (BANet) is proposed. First, due to the diversity of scales and rotation angles of ships, deformable convolution is introduced to build a local attention module (LAM) to better obtain local information of ships and effectively enhance the robustness of the network. Second, a nonlocal attention module (NLAM) is designed to extract the nonlocal features of the SAR image, so as to balance the local features and nonlocal features acquired by the entire network. Finally, the feature pyramid network (FPN) is used to detect ships of different sizes at different scales. The detection results on three datasets demonstrate that the detection precision of our method is higher than that of all comparison methods, and this method achieves the most advanced performance.**

***Index Terms*— Anchor-free mechanism, feature pyramid network (FPN), nonlocal attention, ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) possesses the imaging characteristics of all-weather and all-time. Therefore, it is widely valued by countries all over the world and has become one of the important tools for people to observe the earth. Ship detection in SAR images can detect smuggled and illegal fishing vessels, monitor and manage maritime transport, and improve coastal defense early warning capabilities [1]–[6]. However, due to the uncertainty of sea clutter, the size of ship targets, and the interference of land clutter, the detection performance of ship targets will be greatly affected. Therefore, ship detection is still a meaningful and difficult research direction.

Traditional ship detection methods for SAR images are primarily divided into two steps: sea-land segmentation and target detection. The first step is to separate the land from the SAR image and eliminate land interference for subsequent ship detection. Therefore, the accuracy of segmentation directly affects the result of ship detection. The most classical ship detection algorithm is the constant false alarm rate (CFAR). CFAR can produce a fixed threshold with a constant false alarm probability in a homogeneous background, which can better reduce the disturbance of background noise and clutter. With the continuous improvement of the CFAR algorithm, CFAR detection methods based on statistical distributions, such as K distribution [7] and Gamma distribution [8], have been formed. These methods can produce better detection results for a certain type of specific scene. However, they require artificial analysis of the characteristics of the target and background, and the parameter design is complicated and lacks universality. The detection methods based on CFAR have cumbersome steps and cannot perform end-to-end learning. It is difficult to obtain satisfactory detection results in a complex environment.

With the continuous development of deep learning, a convolutional neural network (CNN) has got remarkable success in optical image detection and classification [9]–[11]. Scholars from various countries have also made a lot of studies on the application of CNN in ship detection for SAR images. The classic target detection methods based on deep networks primarily include one-stage target detection methods and two-stage target detection methods. One-stage target detection algorithms contain single-shot detector (SSD) [12], you only look once (YOLO) series [13], [14], RetinaNet [15], and so on. Chen *et al.* [16] trained a lightweight target detection network based on YOLO, which can reduce the model storage space and ensure higher accuracy at a faster speed. Yang *et al.* [17] constructed a modified one-stage target detection network named R-RetinaNet by using RetinaNet and a rotating bounding box, which solved the mismatch problem of feature scale. Two-stage target detection algorithms mainly include region-CNN (R-CNN) [18], Fast R-CNN [19], Faster

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R-CNN [20], R-FCN [21], and so on. Li *et al.* [22] enhanced the detection performance of Faster R-CNN by adopting a series of strategies such as feature fusion and hard example mining. Lin *et al.* [23] took advantage of Faster R-CNN to construct a ship detection framework, which used a squeeze-and-excitation attention mechanism to strengthen important features. Dai *et al.* [24] designed a feature fusion method by combining bottom-to-up and up-to-bottom features to enhance the positioning and semantic feature of multiscale ships, particularly small ships.

The detection algorithms based on deep learning have high requirements for datasets, so the SAR ship datasets are gradually increasing, such as SSDD [22], HRSID [25], and LS-SSDD-v1.0 [26]. Taking the HRSID as an example, this dataset contains 5604 high-resolution SAR images and 16 951 ships. Small, medium, and large ships account for 54.5%, 43.5%, and 2% of all ships, respectively. The bounding box areas of small, medium, and large ships account for 0∼0.16%, 0.16%∼1.5%, and above 1.5% of SAR images, respectively. It can be seen that HRSID has the characteristics of small and sparse targets. Experiments on this dataset show that the AP_{50} of RetinaNet is only 84.8%, that of Faster R-CNN is only 86.7%, and that of HRSDNet [25] is 89.3%. Although the classical deep learning detection algorithms have achieved certain results, it is still not satisfactory. One of the reasons is the multiscale and multiangle characteristics of the ship itself. Therefore, how to learn the local characteristics of ships more effectively is worth exploring. Another key problem is that ships are sparse and vulnerable to land clutter and sea clutter in large scenes. The commonly used detection algorithms often only focus on the local features of ships and ignore the relationship between image contexts, which makes it impossible to effectively extract the semantic features of the image.

Through the above analysis, we combine the advantages of local and nonlocal attention to construct a new balance attention network (BANet) for anchor-free ship detection in SAR images. First, considering the multiangle and multiscale characteristics of ship targets, the local attention module (LAM) is built to obtain the local characteristics of the ship. Then, a nonlocal attention module (NLAM) is constructed after LAM to extend the receptive field of the network and obtain a better ship detection effect. The most contributions of this article are reflected as follows.

1) In this article, an anchor-free strategy is used to replace the anchor-based method commonly used in ship detection algorithms for SAR images, which eliminates the influence of the setting of anchors on the detection result and improves the detection effect.
2) We propose a new multiscale ship detection method in SAR images on the basis of BANet, which organically integrates local and nonlocal attention mechanisms to acquire a better target detection effect. The experimental results demonstrate that our method has a higher detection accuracy and better generalization performance on the HRSID, SSDD, and LS-SSDD-v1.0 datasets.
3) In this article, deformable convolution is introduced in BANet, and the LAM is constructed to effectively capture the local information of ship targets through learning offset. Moreover, an NLAM is constructed to balance the attention features of the entire network.

The remaining parts of this article are arranged as follows. Section II introduces the related works. Section III presents the BANet method. Section IV gives the experimental results on three datasets and the relevant analysis. Finally, conclusions are made in Section V.

II. Related Works

A. Anchor-Free Strategy

In order to achieve a better detection effect, the commonly used target detection networks usually use the anchor mechanism. First, a group of fixed reference anchors with different sizes and different positions is preset. Then, each reference anchor is responsible for detecting targets whose intersection over union (IOU) is higher than a certain threshold. Finally, a nonmaximum suppression (NMS) algorithm is performed to filter out superfluous reference boxes to acquire the final detected anchors. The anchor mechanism replaces the traditional multiscale traversal sliding window and improves the detection precision and speed. However, the effect of ship detection is limited by the number, aspect ratio, and shape of anchors. As a result, the calculation is redundant, and the performance of the detector is difficult to be further improved.

In recent years, scholars began to study anchor-free detection algorithms to balance the relationship between detection accuracy and detection efficiency. Zhou *et al.* [27] proposed CenterNet for target detection. This method used key point estimation to estimate the center point and used the center point to return all other target attributes. It achieves better detection results than the methods based on anchor mechanism. Cui *et al.* [28] designed a ship detection framework based on spatial shuffle-group enhance (SSE) attention. This algorithm adopted the SSE attention mechanism in CenterNet, which can obtain stronger semantic features of the image while suppressing noise. Kong *et al.* [29] were inspired by the fovea structure of the human eyes and proposed the FoveaBox framework. This framework combined the possible location of the central region of the predicted target and the candidate box of each valid location, and used multiple levels of features to detect targets of different scales. Tian *et al.* [30] proposed the FCOS structure. This structure performed target detection through pixel-by-pixel prediction and used multilayer prediction to increase the recall rate and suppress the fuzzification caused by overlapping predicted boxes.

B. Attention Mechanism

Usually, target detection algorithms mainly focus on local attention. For example, Woo *et al.* [31] combined the advantages of the channel and spatial attention to construct the convolutional block attention module (CBAM), which can refine features and highlight important information.

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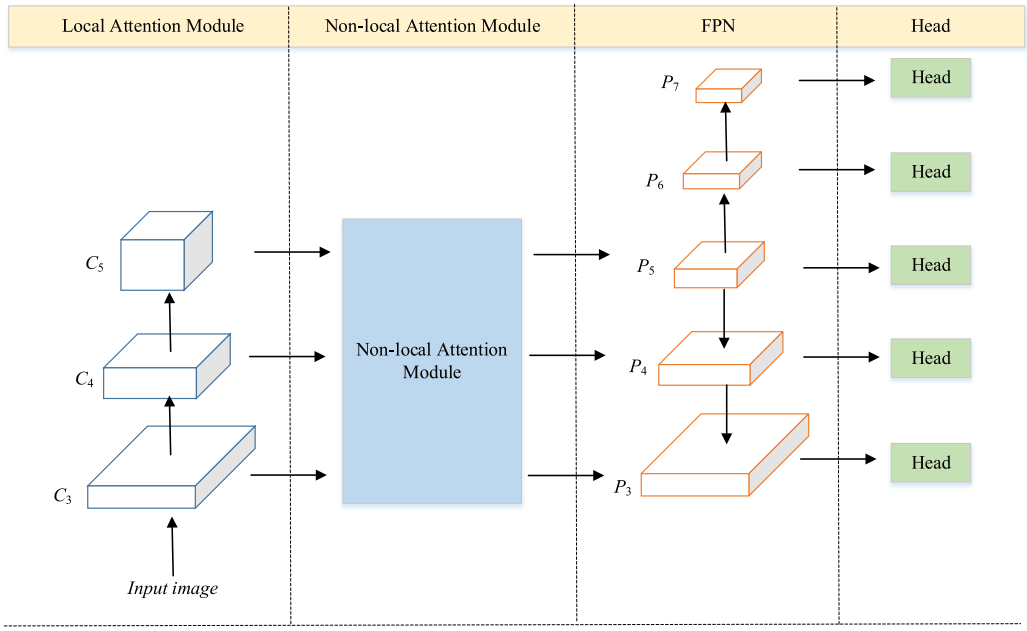


Fig. 1. Overall structure of BANet. It consists of an LAM, an NLAM, an FPN, and detection heads.

Jie *et al.* [32] constructed squeeze-and-excitation networks (SENets) that can enhance helpful features and suppressed features that are not very helpful according to the importance of each feature channel. Li *et al.* [33] proposed selective kernel networks (SKNets). It performed an attention mechanism on the convolution kernel. That is, the network can choose the appropriate convolution kernel by itself to adjust the size of its receptive field. Although SKNet can adaptively adjust the size of the receptive field, it still cannot effectively acquire the local variable features of the target.

Moreover, the aforementioned methods only pay attention to local information and ignore context information. Inspired by traditional image processing, scholars gradually introduced nonlocal attention mechanisms into deep learning algorithms. For example, Wang *et al.* [34] used nonlocal operation to explore long-distance dependence and improved the results of object detection and segmentation. Mou *et al.* [35] proposed a block-based nonlocal attention model to recover each block by gathering helpful information from the entire image. Although the SAR image scene is unknown, it contains a large number of repetitive scenes, such as ocean and land. Therefore, we believe that nonlocal operations can reduce the interference of sea clutter and land clutter, and extract more useful information.

Inspired by the above observations, we explore and improve local attention and nonlocal attention, and propose an anchor-free detection method of ship target in SAR images based on BANet. The specific scheme is described in detail in Section III.

III. Proposed Method

In order to realize the effective detection of ships in SAR images, starting from the characteristics of SAR images, a BANet based on an anchor-free mechanism is constructed in this article. The overall architecture of BANet is designed in Fig. 1. It is composed of an LAM for local feature extraction, an NLAM for context information extraction, a feature pyramid network (FPN) for multiscale prediction, and detection heads for final prediction. The main components of BANet are described in detail in the following.

A. Local Attention Module

The backbone network of the classic anchor-free detection algorithm is built by ResNet. ResNet solves the problem of difficult training of deep CNN models through residual learning and gets grand success in target detection for optics and SAR images. However, the structure of the classic convolution is fixed, which makes it unable to adapt to the deformation of the shape and angle of the ship in the SAR images. Consequently, in this article, we introduce deformable convolution [36] to replace the traditional convolution kernel on the basis of ResNet-101 to build an LAM. For any image change, LAM can learn the offsets of convolution kernel to adapt to different changes in ship targets. The essence of deformable convolution is to add offsets to each convolution sampling point to facilitate the extraction of deformable local features, as shown in Fig. 2. Fig. 2(a) shows a conventional 3×3 convolution window. The blue dots in Fig. 2(b) are the result of adding offsets to the original convolution window. The green dots in Fig. 2(c) represent the receptive field of the conventional convolution kernel on the ship, and the blue dots in Fig. 2(d) represent the receptive field of the deformable convolution on the ship. It can be seen that the sampling position of the deformable convolution is more in line with the shape and size of the ship itself.

Assume that a traditional convolution kernel contains K sampling positions, ω_k and p_k represent the weight of the kth position and the predetermined offset, and $x(p)$ and $y(p)$

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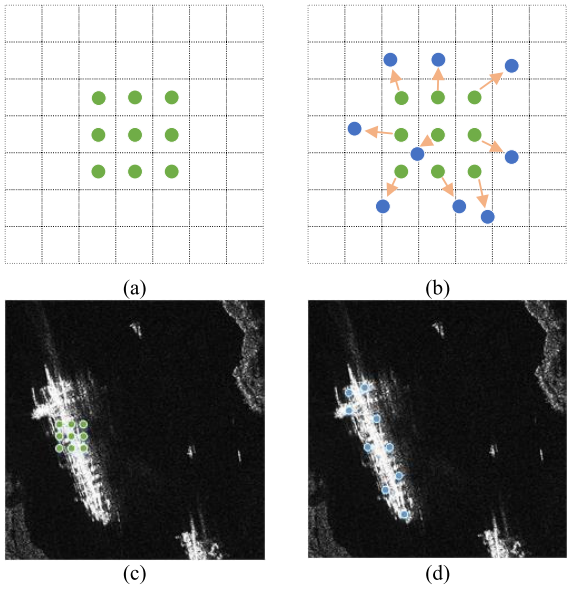


Fig. 2. Schematic of the offsets in deformable convolution. (a) Conventional 3×3 convolution window. (b) Result of adding offsets to the original convolution window. (c) Receptive field of the conventional convolution kernel on the ship. (d) Receptive field of the deformable convolution on the ship.

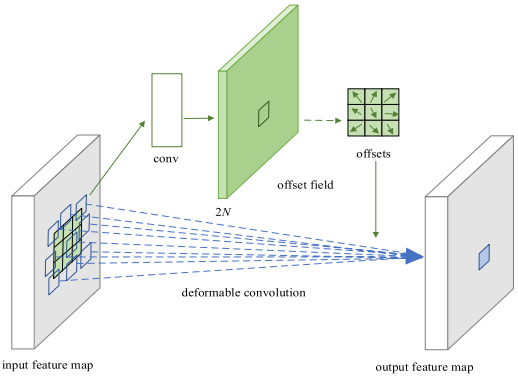


Fig. 3. Structure diagram of deformable convolution.

represent the features of the position p in the input feature map x and the output feature map y. We take the 3×3 ordinary convolution in Fig. 2(a) as an example, where $K = 9$ and $p_k \in \{(-1,-1),(-1,0),\ldots,(1,1)\}$. For each output $y(p)$, nine positions must be sampled from x. These nine positions spread from the center position $x(p)$ to the surroundings. $(-1, -1)$ represents the upper left corner of $x(p)$, and $(1, 1)$ represents the lower right corner of $x(p)$. Therefore, the traditional 3×3 convolution output is

$$y(p) = \sum_{k=1}^{K} w_k \cdot x(p + p_k). \tag{1}$$

Modulation deformable convolution can be expressed as follows:

$$y(p) = \sum_{k=1}^{K} w_k \cdot x(p + p_k + \Delta p_k) \cdot \Delta n_k \tag{2}$$

where Δp_k and Δn_k represent the adaption offset and the modulating factor of the kth position. Δn_k can adjust the input feature amplitude from different spatial positions. It is obtained through a convolutional layer. The range of Δn_k is $[0, 1]$, and the number of channels is N. The offset Δp_k of deformable convolution is acquired by applying a separate convolution module on the input x. As illustrated in Fig. 3, another 3×3 convolutional module is constructed in the model to learn the offset Δp_k. The output of this convolutional module is the offsets at location $x(p)$. Its dimension is the same as x, and the number of channels is $2N$.

B. Nonlocal Attention Module

Ships in SAR images are sparse and easily interfered with by land clutter and sea clutter in large scenes. Deep neural networks usually learn features based on local receptive fields or global information but ignore the correlation between nonlocal blocks in the image. Nonlocal operations were initially applied to nonlocal mean (NLM) filtering [37]. The core idea of NLM is to use surrounding blocks with nonlocal similarity to perform weighted average denoising on pixels. In short, the NLM operation calculates the correlation between blocks to highlight the commonality (region of concern) and eliminating the difference. The deformable convolutional network obtains more accurate local features by learning the offset but does not pay attention to the connection between nonlocal regions. Therefore, we construct an NLAM that weighs all pixels according to the correlation between each pixel. Areas with significant weight are saliency target areas that we want. NLAM can reduce the interference of sea clutter and land clutter, and extract more useful information. The network framework of NLAM is given in Fig. 4.

In Fig. 4, C_l ($l = 3, 4, 5$) represents the output of each level of the backbone network, which is also the input of NLAM. First, the channel compression is performed on C_l through the 1×1 convolutional layer. Then, the result is added with the $2\times$ upsampling results of the $l+1$ level feature map H_{l+1}, and the fused feature map H_l of high-level semantic information and low-level details is obtained (the element addition operation is not performed when $l = 5$). Finally, the output result P_l containing nonlocal features is obtained by linearly weighted integration operation on the information of H_l. The whole process is given as follows:

$$H_l = \begin{cases} \text{Conv}_{1\times1}(C_5), & l = 5 \\ \text{Upsample}(H_{l+1}) + \text{Conv}_{1\times1}(C_l), & l = 3, 4 \end{cases} \tag{3}$$

$$P_l = \text{NL}(H_l),\ l = 3, 4, 5 \tag{4}$$

where $\text{Conv}_{1\times1}$ represents the 1×1 convolution and $\text{NL}(\cdot)$ represents the nonlocal operation. In the nonlocal operation, H_l is the input feature map. h_i and h_j represent the pixel values at position i and position j on H_l. z_i represents the pixel value at position i of the intermediate output image. The nonlocal operation is expressed as

$$z_i = f(h_i, h_j) \cdot g(h_j) \tag{5}$$

where i represents a certain position of the feature map and j represents the index of all possible positions. g represents a univariate linear function used for information transformation. f is the weight calculation rule, which is used to measure

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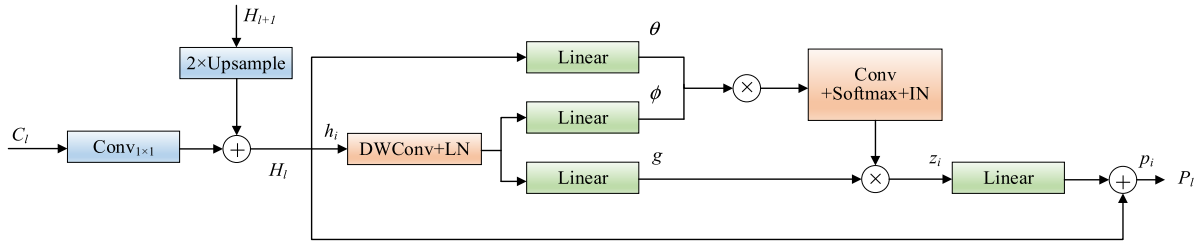


Fig. 4. NLAM.

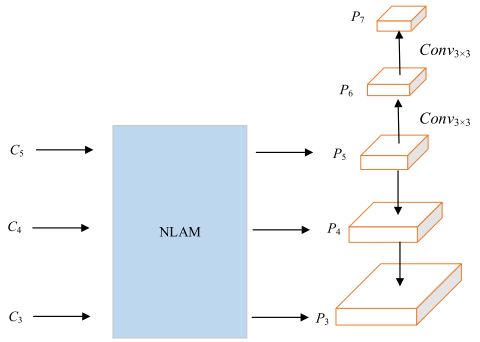


Fig. 5. Structure of FPN.

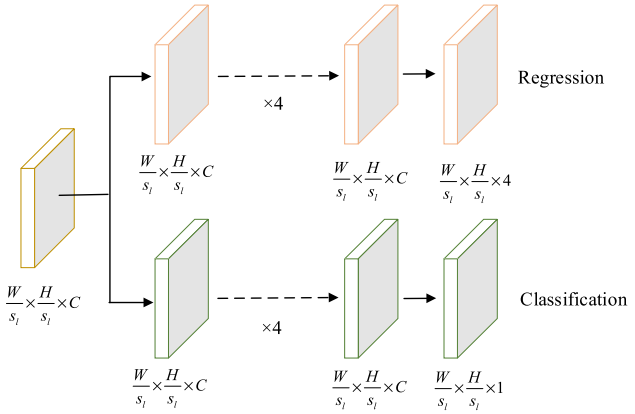


Fig. 6. Framework of the detection head.

the relativity between the ith position and all other positions. $g(h_j)$ is calculated as follows:

$$g(h_j) = W_g \cdot \alpha(h_j) \tag{6}$$

where W_g is the learned weighted matrix. $\alpha(\cdot) = \text{LN}(\text{DWConv}(\cdot))$, where $\text{DWConv}(\cdot)$ denotes the depthwise convolution and $\text{LN}(\cdot)$ denotes the layer normalization. Memory can be compressed through $\text{DWConv}(\cdot)$ and $\text{LN}(\cdot)$.

f is calculated as follows through the dot product:

$$f(h_i, h_j) = \beta\left(\theta(h_i)^T \cdot \phi(h_j)\right) \tag{7}$$

where $\theta(h_i) = W_\theta h_i$ and $\phi(h_j) = W_\phi \cdot \alpha(h_j)$ are linear projections. We use $\beta(\cdot) = \text{IN}(\text{Softmax}(\text{Conv}(\cdot)))$, where $\text{IN}(\cdot)$ denotes instance normalization, so that the attention function can not only rely on all data but also focus on the information of different representation subsets at different locations.

Finally, the final nonlocal attention feature is output through linear operation and residual connection. The calculation formula is given as follows:

$$p_i = W_z z_i + h_i \tag{8}$$

where z_i is calculated by formula (5) and h_i represents the residual connection. p_i represents the pixel value at position i of the output feature map P_l.

C. FPN

Considering the multiscale problem in ship detection for large scene SAR images, we bring an FPN [38] in BANet. The feature maps obtained by the image through the two modules of local attention and nonlocal attention form a feature pyramid. The structure of FPN is illustrated in Fig. 5, which is primarily consisted of P_3, P_4, P_5, P_6, and P_7. The feature maps P_3, P_4, and P_5 are obtained through NLAM. P_6 is generated by the convolution of P_5 with a step of 2. P_7 is generated by the convolution of P_6 with a step of 2, which is mainly used to enhance the detection effect of large targets. P_l is calculated as follows:

$$P_l = \begin{cases} \text{NLAM}(C_l), & l = 3, 4, 5 \\ \text{Conv}_{3\times3}(P_{l-1}), & l = 6, 7 \end{cases} \tag{9}$$

where $\text{Conv}_{3\times3}$ represents the 3×3 convolution. $\text{NLAM}(\cdot)$ represents the nonlocal attention operation, and $\text{NLAM}(C_l) = \text{NL}(H_l)$. Through the combination of up-to-bottom feature maps, FPN can obtain comprehensive feature maps by fusing high-layer feature maps with plentiful semantics and low-layer feature maps with higher resolution. Therefore, the network can maintain a higher resolution while extracting semantic information and can focus on the relationship between different positions of the feature maps and highlight the important regions. At the same time, FPN can use different feature layers to capture features of ships of different sizes, so as to achieve multiscale target detection.

D. Detection Head

In order to complete the final prediction, we use the detection head shown in Fig. 6. This network contains two specific subtasks, classification and regression, both of which are composed of five 3×3 convolutional layers. The classification task is applied to predict the probability of the existence of the target, and the regression task is applied to predict the location of the detection box. The input of the detection head is the

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output P_l ($P_l \in \mathbb{R}^{(W/s_l)\times(H/s_l)\times C}$) of FPN, where $l = 3, 4, 5, 6, 7$, W and H represent the height and width of the input feature map, C indicates the channel dimension of P_l (we use C as 256), s_l represents the downsampling scale corresponding to the input feature map, and $s_l = 2^l$.

To perform classification and regression, we need to find the location on the original SAR image corresponding to each position (x, y) on P_l. ($x = 0, 1, \ldots, W/s_l - 1$ and $y = 0, 1, \ldots, H/s_l - 1$). The original image position corresponding to the position (x, y) on P_l is defined as (x', y'), where $x' = s_l(x + 0.5)$ and $y' = s_l(x + 0.5)$. In this article, we use $d' = (d'_l, d'_t, d'_r, d'_b)$ to locate the bounding box, where d'_l, d'_t, d'_r, and d'_b, respectively, indicate the distance from the position (x', y') to the four sides of the box. If the position (x', y') is in the bounding box $B = (x_0, y_0, x_1, y_1)$ of the ground-truth image, the pixel at this location is classified as a positive sample. We denote the relevant class label as $\hat{c} = 1$. Alternatively, it is classified as a negative sample, and the relevant class label is expressed by $\hat{c} = 0$. For positive samples, the regression box $\hat{d} = (\hat{d}_l, \hat{d}_t, \hat{d}_r, \hat{d}_b)$ can be expressed as

$$\begin{cases} \hat{d}_l = x' - x_0, & \hat{d}_t = y' - y_0 \\ \hat{d}_r = x_1 - x', & \hat{d}_b = y_1 - y' \end{cases} \tag{10}$$

where (x_0, y_0) and (x_1, y_1) represent the coordinates of the upper left and lower right corners in the bounding box of the ground-truth image, respectively. The anchor-based detector detects targets of different feature levels by setting anchors of different sizes. The anchor-free detection methods perform target detection by limiting the bounding box regression range of each level, that is, different pyramid levels detect ships of different scales. $[m_{l-1}, m_l]$ is defined as the scale boundary of level l for regression. If a position satisfies $\max(\hat{d}_l, \hat{d}_t, \hat{d}_r, \hat{d}_b) > m_l$ or $\max(\hat{d}_l, \hat{d}_t, \hat{d}_r, \hat{d}_b) < m_{l-1}$, the pixel at that position is set as a negative sample. In this article, m_2, m_3, m_4, m_5, m_6, and m_7 take 0, 64, 128, 256, 512, and ∞, respectively. Experiments show that the multilevel prediction increases the identification degree of the bounding box so that the detection result can reach the same level as the anchor-based detector.

E. Loss Function

During the training period, our loss function is represented as follows:

$$L = \frac{1}{N_{\text{pos}}} \sum_l \sum_{x,y} \left\{ L_{\text{cls}}\left(c^l_{x,y}, \hat{c}^l_{x,y}\right) + \left[\hat{c}^l_{x,y} > 0\right] L_{\text{reg}}\left(d^l_{x,y}, \hat{d}^l_{x,y}\right) \right\} \tag{11}$$

where L_{cls} is the classification loss. To eliminate the influence of category imbalance, we adopt the focus loss function proposed in [15] when training the classification network. L_{reg} represents the regression loss. We apply the IOU loss function presented in [39] when training the regression network. l represents the level of the feature map. N_{pos} is the amount of positive samples. $[\hat{c}^l_{x,y} > 0]$ is the Iverson bracket. If $\hat{c}^l_{x,y} > 0$, then $[\hat{c}^l_{x,y} > 0] = 1$; otherwise, $[\hat{c}^l_{x,y} > 0] = 0$.

IV. Experimental Results and Analysis

We conducted an ablation experiment, a comparison experiment with the advanced algorithms, and a discussion on three different datasets in this section. The details are introduced in the following.

A. Datasets and Training Settings

In order to evaluate the performance of BANet, we use three datasets for experiments: HRSID [25], SSDD [22], and LS-SSDD-v1.0 [26]. The HRSID dataset consists of 5604 original SAR images with a resolution of 0.5, 1, and 3 m, all of which are 800 × 800 in size. These images have different environments, polarizations, and imaging methods. Some examples are shown in Fig. 7(a). Ship targets are marked by experts, and the corresponding optical remote sensing images are used to eliminate interference. The format of Microsoft Common Objects in Context (MS COCO) is used to label each ship, including area and bbox (x, y, width, height). x and y represent the coordinates of the upper left corner of the annotation bounding box, and the width and height indicate the width and height of the bounding box. In our experiment, 65% of the dataset is divided into training data, and the rest 35% is divided into testing data. The images of SSDD are acquired by RadarSat-2, TerraSAR-X, and Sentinel-1. The dataset consists of 1160 SAR images with a resolution of 1–15 m of different sizes, such as 416 × 323 and 501 × 355. Some examples are shown in Fig. 7(b). We use SSDD to verify the generalization performance of detection methods. LS-SSDD-v1.0 are obtained from Sentinel-1, including 15 large scene images with a size of 24 000 × 16.000 and 9000 subimages with a size of 800 × 800. The image resolution is 5 m × 20 m. Some examples are shown in Fig. 7(c). This dataset has relatively small ship targets, and the detection is difficult. We select a large scene image from LS-SSDD-v1.0 to verify the practicability of our method.

The proposed BANet uses ResNet-101 pretrained on ImageNet to initialize the overall network and uses the stochastic gradient descent (SGD) method to train the network. There are eight SAR images in a minibatch. The number of iterations is 180 k. The beginning learning rate is set to 0.001, which decreased ten times at 120k and 160k, respectively. The weight decay is set to 0.0001, and the momentum is 0.9. Training and testing are completed on the PyTorch framework of Ubuntu and the Nvidia Tesla P100 GPU.

B. Evaluation Criteria

We choose precision (Precision), recall (Recall), F1 score (F1), and average precision (AP) as the evaluation indicators to verify the performance of BANet. The definitions of these indicators are given as follows:

$$\text{Precison} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{12}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{13}$$

$$\text{F1} = \frac{2 \times \text{Precison} \times \text{Recall}}{\text{Precison} + \text{Recall}} \tag{14}$$

$$\text{AP} = \frac{1}{101} \sum_{r \in S} \text{Precison}|_{\text{Recall}=r} \tag{15}$$

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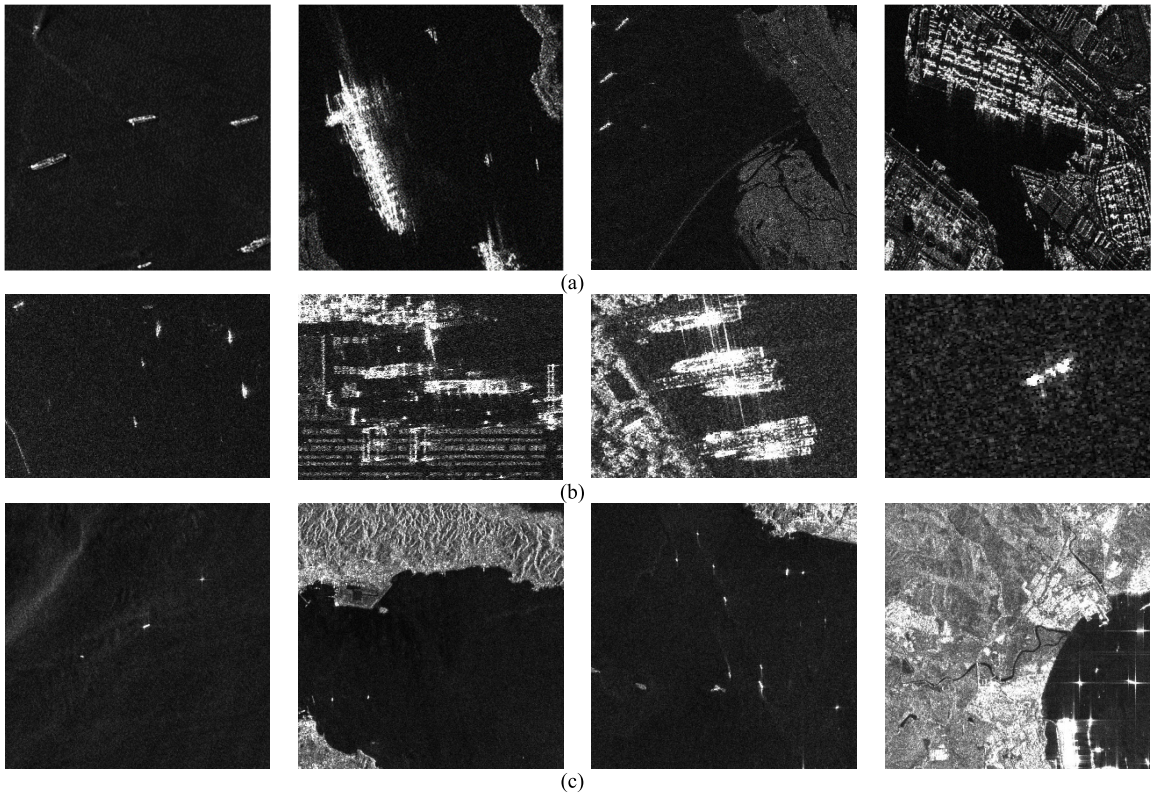


Fig. 7. Some sample images in three datasets. (a) Images in HRSID. (b) Images in SSDD. (c) Images in LS-SSDD-v1.0.

where N_{TP} represents the sum of correctly detected ships, N_{FP} represents the sum of falsely detected ships, and N_{FN} is the sum of missed ships. Precision is the proportion of accurately predicted ships in all prediction. Recall represents the proportion of accurately predicted ships in all ships of ground truth. F1 is defined to judge the overall performance of the network. AP shows the average accuracy of a group of equidistant recall rates $S = [0, 0.01, \ldots, 1]$, which can be drawn to display the PR curve. The average accuracy can be divided into AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l according to specific measurement standards. AP represents the average accuracy when IoU = 0.50:0.05:0.95. AP_{50} represents the average accuracy when IoU = 0.5. AP_{75} represents the average accuracy when IoU = 0.75. AP_s, AP_m and AP_l represent the average accuracy of the small, medium, and large targets. When detecting a ship, if the IOU of the ship's detection result (detection bounding box) and the ground truth (annotation bounding box) is greater than 0.5 (AP_{50} is adopted), we consider that this ship is detected correctly.

In addition, frames per second (FPS) shows the test speed of detection algorithms. The larger the FPS, the faster the algorithm runs. FPS is defined as follows:

$$\mathrm{FPS} = 1/T \tag{16}$$

where T represents the average inference time for processing an image.

C. Ablation Experiment and Model Analysis

We construct an anchor-free detection method based on LAM and NLAM to adapt to the characteristics of ship targets in SAR images. The effect of these two modules is verified in the following.

TABLE I
EVALUATION OF LAM

	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Baseline	0.735	0.944	0.824	0.723	0.865	0.867
ACNet+FPN	0.740	**0.946**	0.833	0.729	0.863	0.875
LAM+FPN	**0.886**	0.939	**0.907**	**0.872**	**0.992**	**0.977**

1) Effect and Analysis of LAM: The proposed algorithm uses deformable convolution to build a local attention backbone architecture that pays more attention to ship targets. To verify the effect of this module, LAM is tested on HRSID and compared with Baseline (ResNet + FPN) and ACNet [40] + FPN. ACNet strengthens the feature extraction capability of the network by designing three convolution kernels. Evaluation indicators AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l are used for effect evaluation, and Table I shows the results. Compared with Baseline, our method has increased by 15.1% and 8.3% in AP_{50} and AP_{75}, respectively, and the AP for ships of different scales has been greatly improved. Moreover, the AP of LAM is much better than that of ACNet. This is mainly because LAM can extract local feature information more effectively, so as to achieve accurate matching of ships. The result demonstrates that the LAM module heightens the performance of ship detection and enhances the accuracy of target positioning.

In order to illustrate the effect of LAM more intuitively, we give the detection effect of an image, as revealed in Fig. 8.

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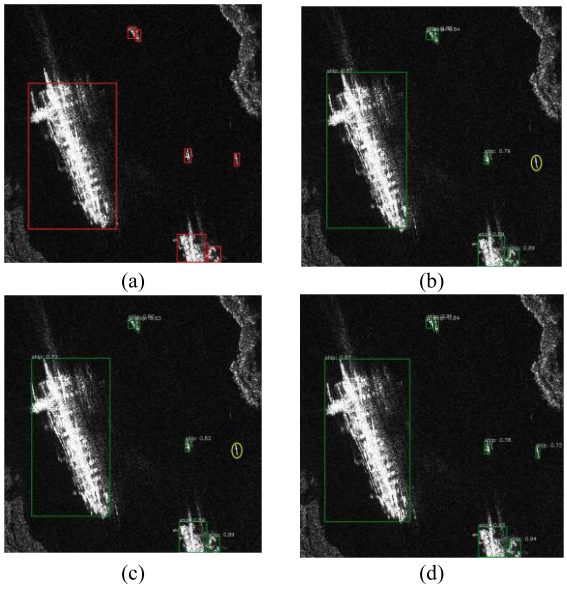


Fig. 8. Test effects of LAM. The red box indicates ground truth, the green box indicates detection results, and the yellow circle indicates missed target. (a) Ground truth. (b) Baseline. (c) ACNet + FPN. (d) LAM + FPN.

TABLE II
EVALUATION OF NLAM

	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Baseline	0.735	0.944	0.824	0.723	0.865	0.867
Baseline+NLN	0.738	0.945	0.831	0.726	**0.868**	0.863
Baseline+NLAM	**0.742**	**0.946**	**0.840**	**0.731**	0.867	**0.877**

Fig. 8(a) displays the ground truth. There is a small ship on the right-hand side in Fig. 8(a), but it is missed (marked by a yellow circle) in Fig. 8(b) and (c). In Fig. 8(d), the LAM proposed in this article correctly detects the small target with a confidence of 0.72. This fully proves that LAM can sufficiently learn the feature of ships and accurately locate smaller targets.

2) Effect and Analysis of NLAM: In this article, NLAM is constructed to further strengthen the connection between contexts and obtain the nonlocal features of the image. NLAM is tested on the HRSID and compared with Baseline (no nonlocal operation) and Baseline + NLN (nonlocal operation is NLN [34]). NLN obtains the correlation between pixels by calculating the relationship between different locations. Table II shows the test results. We can know that, compared with Baseline, AP, AP_{50}, AP_{75}, AP_s, AP_m, and AP_l of NLAM have a certain increase, and AP_{75} has increased by 1.6%. Compared with NLN, AP_{75} of NLAM is increased by 0.9%, which proves that NLAM can effectively use context information to enhance the effect of target detection.

In order to further illustrate the effect of NLAM, we visualize an SAR image, and the result is shown in Fig. 9. Area A (marked by a yellow box) of ground truth in Fig. 9(a) contains three ships. The two ships in the upper left corner of area A in Fig. 9(b) are detected as one. After using NLN to improve, the two ships in the upper left corner of area A in Fig. 9(c) are correctly detected, but one ship in the lower right corner is missed. After using NLAM to improve, three ships are correctly detected in Fig. 9(d). Therefore, NLAM can correctly distinguish targets that are close to each other and increase detection precision.

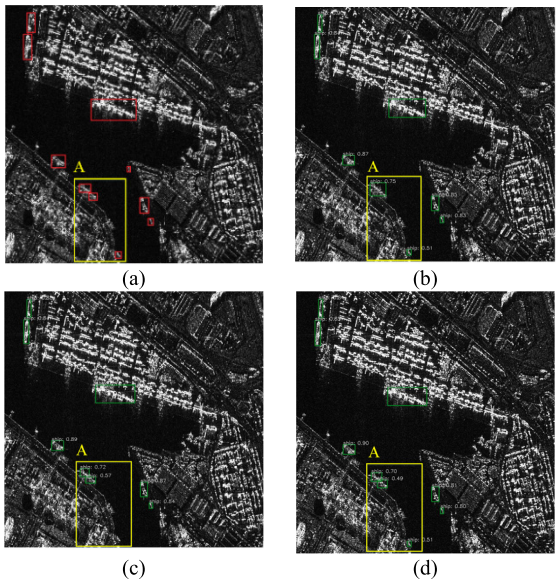


Fig. 9. Verification effects of NLAM. (a) Ground truth. (b) Baseline. (c) Baseline + NLN. (d) Baseline + NLAM.

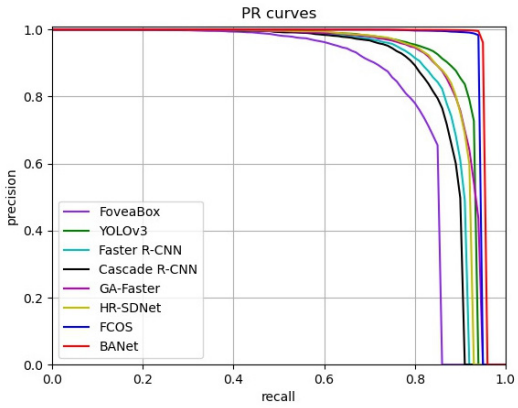


Fig. 10. PR curves of eight detection methods.

D. Comparison Experiment

To verify the validity of BANet, we compare BANet with the classic detection algorithms. These algorithms include the anchor-free target detection algorithm (FoveaBox) proposed in [29], the Darknet-53-based target detection algorithm (YOLOv3) proposed in [41], the detection algorithm (Faster R-CNN) based on region proposal networks (RPNs) proposed in [20], the target detection algorithm (Cascade R-CNN) improved by different IOU thresholds proposed in [42], the detection algorithm (GA-Faster) that uses semantic features to guide the generation of anchor boxes proposed in [43], the high-resolution detection network-based SAR image ship detection method (HR-SDNet) proposed in [44], and the one-stage anchor-free target detection method (FCOS) presented in [30]. Table III shows the experimental results of these eight methods on the HRSID dataset. We can see that BANet

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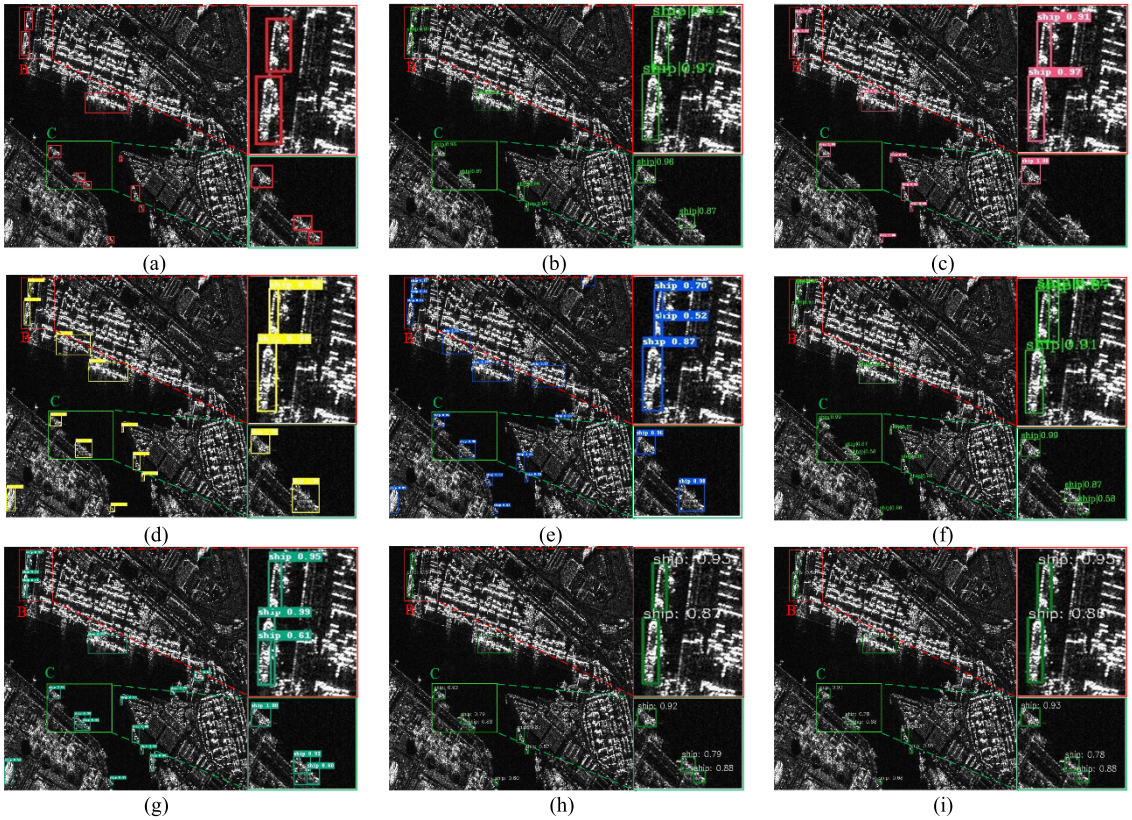


Fig. 11. Comparison of ship detection results of Sample Image I. Red box B and green box C are enlarged areas. (a) Ground truth. (b) FoveaBox. (c) YOLOv3. (d) Faster R-CNN. (e) Cascade R-CNN. (f) GA-Faster. (g) HR-SDNet. (h) FCOS. (i) BANet.

TABLE III
OBJECTIVE EVALUATION OF EIGHT DETECTION METHODS

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}	*FPS*
FoveaBox	0.582	0.635	0.607	0.811	0.650	14
YOLOv3	0.621	0.666	0.643	0.910	0.712	**25**
Faster R-CNN	0.620	0.663	0.641	0.878	0.700	11
Cascade R-CNN	0.639	0.685	0.661	0.865	0.722	11
GA-Faster	0.648	0.695	0.671	0.905	0.738	13
HR-SDNet	0.652	0.695	0.673	0.896	0.748	5
FCOS	0.898	0.909	0.903	0.940	0.917	17
BANet	**0.926**	**0.933**	**0.929**	**0.950**	**0.929**	14

has the highest precision and recall, and the detection speed is moderate. AP_{75} of BANet is 1.2% higher than that of the second-best method FCOS, which illustrates that BANet can get a higher quality bounding box and locate the target more accurately. The corresponding PR curve is drawn in Fig. 10. The area under the red curve corresponding to BANet is the largest, indicating that our method has the optimal performance.

In addition, we randomly select two images in the HRSID dataset to visually verify the proposed model (named the two images as sample and sample, respectively). Figs. 11 and 12 show the detection results. Figs. 11(a) and 12(a) are ground truth, and the remaining subimages are the detection results of eight algorithms.

Ships in Fig. 11 are in an inshore environment. Due to the influence of land scattering, ships are difficult to detect. In this article, we enlarge area B (marked by the red box) and area C (marked by the green box) of each image in Fig. 11 and display them on the right. It can be seen from Fig. 11 that there are two ships in area B, but Cascade R-CNN, GA-Faster, and HR-SDNet are all incorrectly detected as three. Area C contains three ships, but YOLOv3 incorrectly detects as one, FoveaBox, Faster R-CNN, and Cascade R-CNN incorrectly detect as two, and HR-SDNet incorrectly detects as four. Both FCOS and BANet correctly predict these five ships in area B and area C, but the precision and positioning accuracy of BANet is higher than that of FCOS. In summary, BANet can more accurately detect inshore ships and ships that are closer to each other, and has superior performance.

The scales of the ships in Fig. 12 vary greatly, which requires the higher multiscale performance of the detection model. Fig. 12(a) is the ground truth, which contains one large ship and six small ships. FoveaBox, Faster R-CNN, Cascade R-CNN, GA-Faster, and HR-SDNet do not correctly detect the large ship. FoveaBox, YOLOv3, Faster R-CNN, Cascade R-CNN, and GA-Faster all have a certain degree of missing ships. Faster R-CNN and HR-SDNet incorrectly treat land as the target. Both FCOS and BANet detect all ships correctly. However, the confidence level of the large ship detected by FCOS is 0.86, and the confidence level of the large ship detected by BANet is 0.88. Moreover, BANet is more accurate than FCOS for the positioning of a large ship. The confidence level of BANet in two ships approaching each other is also higher than that of FCOS. This shows that BANet can not only

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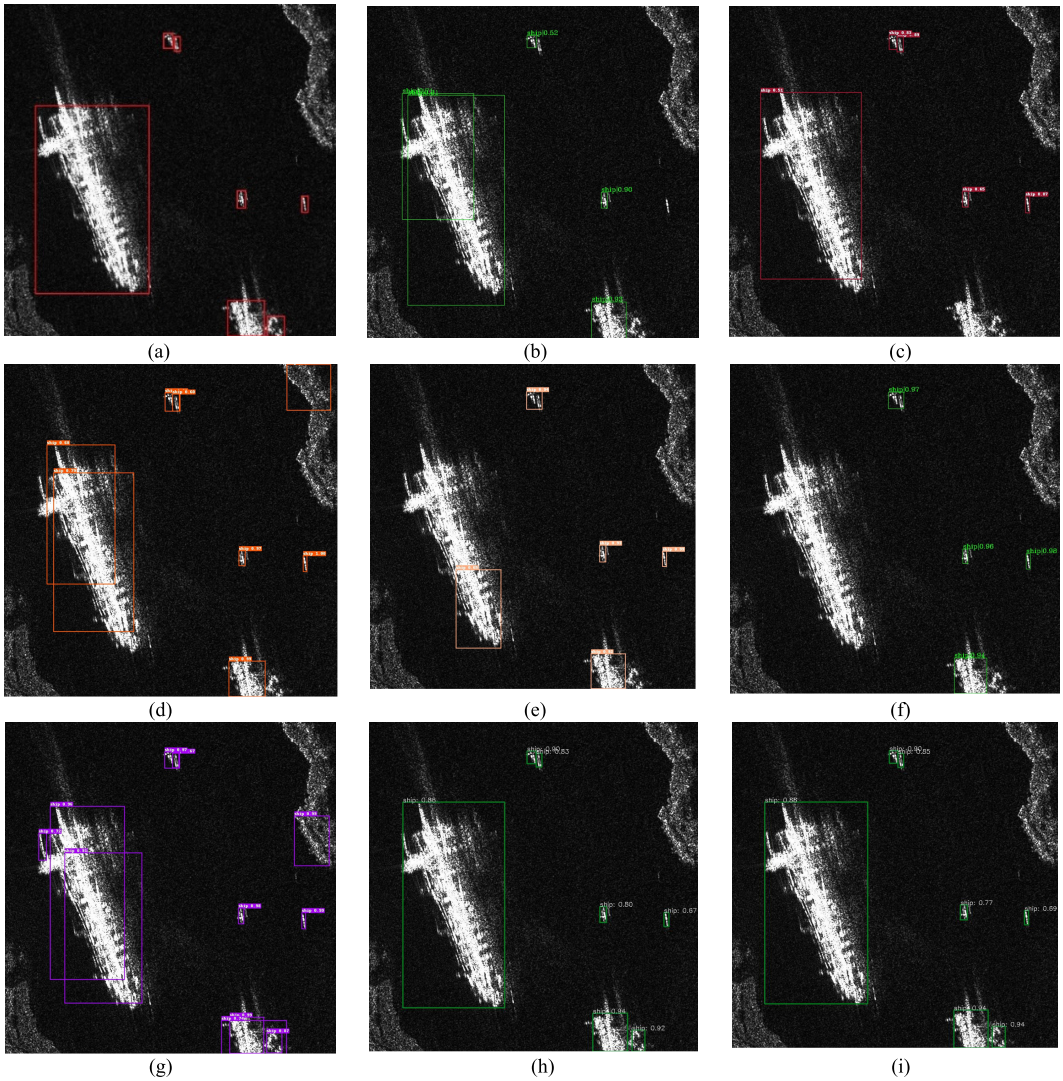


Fig. 12. Comparison of ship detection results of Sample Image II. (a) Ground truth. (b) FoveaBox. (c) YOLOv3. (d) Faster R-CNN. (e) Cascade R-CNN. (f) GA-Faster. (g) HR-SDNet. (h) FCOS. (i) BANet.

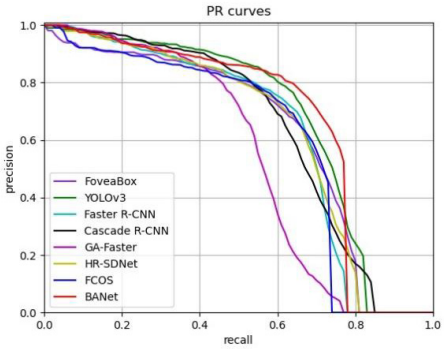


Fig. 13. PR curves of generalization experiments of different detection methods.

accurately detect multiscale targets but also locate targets more exactly.

E. Generalization Experiment

To test the generalization performance of BANet, we use the HRSID dataset for training and then directly tests on the SSDD dataset. Table IV shows the experimental results. We can see that BANet has the highest Precision and F1. Recall of BANet is slightly lower than that of Cascade R-CNN, and AP_{50} of BANet is slightly lower than that of YOLOv3, but BANet obtains the optimal value on the more stringent indicator AP_{75}. The corresponding PR curve is drawn in Fig. 13, and BANet has the largest area under the curve. In general, BANet has better robustness and stronger generalization performance.

TABLE IV
OBJECTIVE EVALUATION OF GENERALIZATION EXPERIMENTS OF DIFFERENT DETECTION METHODS

	Precision	*Recall*	*F1*	AP_{50}	AP_{75}
FoveaBox	0.206	0.333	0.255	0.626	0.054
YOLOv3	0.245	0.358	0.291	**0.678**	0.104
Faster R-CNN	0.189	0.308	0.234	0.625	0.054
Cascade R-CNN	0.222	**0.369**	0.277	0.637	0.086
GA-Faster	0.186	0.297	0.229	0.539	0.072
HR-SDNet	0.196	0.329	0.246	0.632	0.062
FCOS	0.216	0.318	0.257	0.607	0.091
BANet	**0.251**	0.355	**0.294**	0.673	**0.113**

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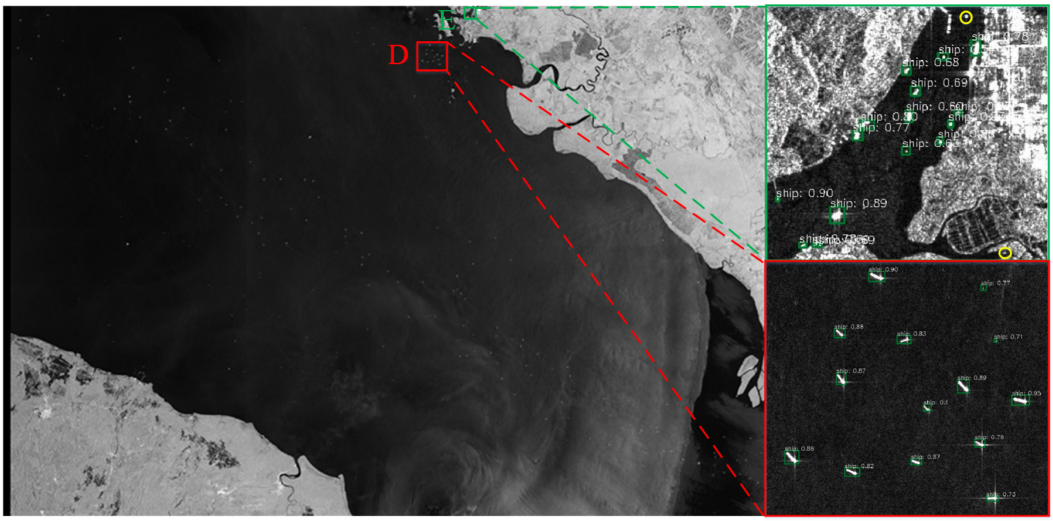


Fig. 14. Detection result of SAR images in the large scene. The green box indicates detection results, and the yellow circle indicates missed target.

Comparing the PR curves in Figs. 10 and 13, in the generalization experiment, we can find that the performance of all methods has decreased, which may be caused by differences in image resolution or sea conditions, and so on. As a result, this means that the generalization performance of the SAR image ship detection algorithm is worthy of further study.

F. Verification on Large Scene SAR Image

In consideration of the practical application of the SAR image ship detection algorithm, a large scene SAR image in the LS-SSDD-v1.0 dataset is used for experiments. First, we use 6000 subimages in the LS-SSDD-v1.0 dataset for training and use 600 subimages for testing and then stitch the test results into a 24 000 $\times$ 16000 SAR image. The final result is shown in Fig. 14. We enlarge area D (marked by the red box) and area E (marked by the green box) and display them on the right-hand side. It can be seen from Fig. 14 that all the ships on the broad sea in area D are detected correctly. In area E, the ships are densely arranged and in an inshore environment. The detection of area E is difficult, but the proposed method still performs well. This shows that the proposed BANet is robust and can achieve satisfactory detection results on different datasets.

V. Conclusion

Taking into account the characteristics of SAR image scenes and ship targets, we present an anchor-free method for SAR image multiscale ship detection based on BANet. Deformable convolution is introduced to construct LAM, which can extract the local features of ships with effect and make the model more suitable for multiangle and multiscale targets. Furthermore, NLAM is designed to balance the local and nonlocal features of the entire network, and context information is used to better extract image semantic information. Finally, FPN is constructed for multiscale target detection. The experimental results demonstrate that BANet enhances the detection precision of multiscale targets and improves the detection effect of inshore ships and ships that are closer to each other in SAR images. In the future, the generalization performance of this method will be further studied and improved to obtain better detection results.

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